DHRUVAL JAVIA

Mechanical engineering graduate student with two years of industry experience, passionate about building practical, scalable solutions for energy challenges, with an innovation and sustainability-driven outlook. Experienced in thermal and electrochemical system modeling, design optimization, and simulation-based analysis of batteries and industrial energy systems.

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EDUCATION

Year	Institute	Qualification	CGPA
2024 – present (Expected 06/2026)	Stanford University	M.S. in Mechanical Engineering (Depth in Energy and Transport Sciences)	3.731/4
2018-2022	Indian Institute of Technology, Bhubaneswar	B. Tech. in Mechanical Engineering	9.39/10

WORK EXPERIENCE

THERMAX LTD. | Trainee – Advent Rotational Program | August 2022 – July 2024

- *Mapped manufacturing processes for boilers and heaters on the factory floor, identified bottlenecks, and proposed automation improvements*; prepared detailed **RLA** and **RCA** reports based on site evaluations.
- Led a team project on **standardization of the design and techno-commercial offering of a biomass-fired thermal oil heater**, leading to a **7.4% reduction in total weight** and a **10.2% decrease in total footprint area** as compared to the previous design. Responsibilities included *grate & BOP sizing, layout preparation, thermal & pressure drop simulations, and efficiency calculations.*
- As part of the project, developed a **thermal model** of the **APH**, **Economizer** and **MPA furnace** in **Excel** with **VBA automation** *to iteratively design the heater*, and *validated it with existing data*. The model provided *inlet/outlet temperature values* within **2%**, **6%** and **10% error** in APH, Economizer and MPA furnace respectively.

ACADEMIC PROJECTS

EFFICIENT BATTERY THERMAL MANAGEMENT SYSTEMS FOR EV | COMSOL, Electrochemical Modelling
2021-2022 | Final Year B. Tech. Thesis Project | Supervisor: Dr. B. R. Pattabhi

- Performed degradation studies through **multiphysics numerical simulations** using **COMSOL** software by employing a **coupled 1-D electrochemical and 2-D thermal model of the battery pack** integrated with a **capacity fade model**.
- Published the work in the **"Energy Storage"** journal having an *impact factor of* **3.2**.

MACRO-SCALE DESIGN ASPECTS OF EV BATTERY PACK | MATLAB, Excel, Battery Pack Design
2021 | Summer Research Internship Project | Supervisors: Dr. Sundararajan Natarajan (IIT Madras) and Dr. B. R. Pattabhi (IIT Bhubaneswar)

- *Performed battery pack design calculations* by developing a system of equations in **Excel** and implementing it in a **MATLAB app**. The app sliders were used to *analyze the effect of battery pack design parameters on EV performance.*

AEROCASE | Sheet Metal Working, TIG Welding, Sand Casting, Woodworking, 3D Printing, Fusion 360
2025 | Individual Course Project | ME 203: Design and Manufacturing, Stanford University

- *Designed* and *fabricated* a compact, innovative, and functional instrument case that ensures the safe storage of Aerodrum instruments while reflecting the aesthetic essence of music.
- Fabricated using *Al 5052 aluminum sheet, A356 cast aluminum, pine wood, Duron,* and *polyurethane foam.*

GAS DYNAMICS SIMULATOR | MATLAB, Vector Algebra, Hash Grid Optimization
2025 | Individual Personal Summer Project

- Built a **MATLAB-based simulator** for *particle dynamics and association chemical reactions in an ideal gas*, implementing **hash grid optimization** to *reduce collision detection algorithm complexity*.
- Developed an interactive MATLAB app with *numeric and GUI-based input to improve user experience*, providing time-series plots and dynamic animations as post-processing outputs for analysis.

TECHNICAL SKILLS

CAD & Design: SolidWorks, Fusion 360 | **Simulation & Analysis:** COMSOL, ANSYS, MATLAB
Programming: Python, Arduino, VBA | **Manufacturing:** Sheet metal, TIG welding, Sand casting, Woodworking, 3D printing

LEADERSHIP & ACHIEVEMENTS

- **Student Placement Coordinator** (2022) – Coordinated with recruiters and institute stakeholders.
- Led 4-member team in IIT Bombay e-Yantra Robotics Competition (2019–20).
- **5th rank** (82 participants), GrabCAD Shop Challenge – *Designed 3D printable 2-in-1 mechanical hand blender*